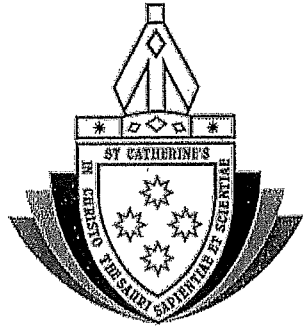




Student Number: _____

St. Catherine's School
Waverley

August 2010

TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics Extension 1

General Instructions

- Reading Time – 5 minutes
- Working Time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Complete each section in a separate booklet

- Attempt Questions 1 – 7
- All questions are of equal value
- Questions to be presented in Sections:

Booklet 1 – Questions 1-2
Booklet 2 – Questions 3-4
Booklet 3 – Questions 5-6
Booklet 4 – Question 7

- Total Marks – 84

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x, x > 0$

Total marks -120

Attempt Questions 1-10

All questions are of equal value

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

Question 1 (12 marks) (Use Writing Booklet 1)

Marks

- (a) Differentiate $\tan^{-1} \frac{x}{2}$ 1
- (b) Given that $\cos \alpha = \frac{5}{13}$ find the value of $\cos 2\alpha$ 2
- (c) Consider the cubic equation $x^3 - 7x - 6 = 0$. If two roots of this equation are -1 and 3 , find the third root. 1
- (d) Find $\frac{d}{dx}(x^2 e^{-x^2})$ 1
- (e) The acute angle between the lines $y = (m+2)x$ and $y = mx$ is 45°
- (i) Show that $\left| \frac{2}{m^2 + 2m + 1} \right| = 1$ 1
- (ii) Hence find the possible values for m 2
- (f) Use the substitution $u = 1 + \ln x$ to evaluate 3

$$\int_1^e \frac{1}{x} (1 + \ln x)^3 dx$$

Question 2 (12 marks) (Use Writing Booklet 1)

Marks

- (a) The variable point $P(t+1, 2t^2+1)$ lies on a parabola. 2
Find the Cartesian equation of the parabola.
- (b) Solve the equation $\frac{2x+3}{x-4} \leq 1$ 2
- (c) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 3x}{5x}$ 2
- (d) Evaluate $\int_0^\pi 2 \cos^2 \frac{x}{4} dx$ 2
- (e) Find the coefficient of x^5 in the binomial expansion of $\left(x^2 + \frac{2}{x}\right)^{10}$ 2
- (f) $A(-3,4)$ and $B(1,2)$ are two points. Find the coordinates of the point (x,y) 2
which divides the interval AB externally in the ratio $3:1$

Question 3 (12 marks) (Use Writing Booklet 2)**Marks**

- (a) Consider the function $f(x) = \frac{x-2}{x-1}$
- (i) Show that the function is increasing for all values of x in the function's domain. 2
- (ii) Sketch the graph of the function showing clearly any intercepts on the coordinate axes and the equations of any asymptotes. 2
- (iii) Find the equation of the inverse function $f^{-1}(x)$ 1
- (iv) Deduce, from your result in (iii), that the graph of the function $f(x)$ is symmetrical about the line $y = x$ 1
- (b) Consider the function $y = \frac{1}{2}\cos^{-1}(x-1)$
- (i) Find the domain and range of the function. 2
- (ii) Sketch *neatly* the graph of the function, showing clearly the coordinates of the end points. 1
- (iii) The region in the first quadrant bounded by the curve $y = \frac{1}{2}\cos^{-1}(x-1)$ and the coordinate axes is rotated through 360° about the y axis.
- Find the volume of the solid of revolution, giving your answer in simplest exact form 3

Question 4 (12 marks) (Use Writing Booklet 2)**Marks**

- (a) Consider the equation $4e^{-x} - \tan x + 1 = 0$ which has a root $x = \alpha$
- (i) Show that $1 < \alpha < 1.5$ 1
- (ii) Using $x = 1$ as a first approximation of the root use one application of Newton's method to find a better approximation of this root. 3
- Write your answer correct to 4 significant figures
- (b) The rate of growth of a bacteria colony is proportional to the excess of the colony's population over 5000 and is given by
- $$\frac{dN}{dt} = k(N - 5000) \text{ where } k \text{ is a positive constant and } t \text{ is the time in days}$$
- (i) Show that $N = 5000 + Ae^{kt}$ is a solution of the above differential equation. 1
- (ii) If the initial population is 15000, and it reaches 20000 after 2 days, find the value of A and k 3
- (iii) Hence, calculate the expected population after a further 5 days. 1
- (c) A committee of 3 women and 7 men are to be seated randomly at a round table
- (i) What is the probability that the three women are seated together. 1
- (ii) The committee elects a President and a Vice President. What is the probability that they are seated opposite one another. 2

Question 5 (12 marks) (Use Writing Booklet 3)

Marks

- (a) Use mathematical induction to prove to prove that for all $n \geq 2$

4

$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{(n-1)}{n!} = 1 - \frac{1}{n!}$$

- (b) Show that $\frac{d^2x}{dt^2} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$

2

- (c) A particle moves in a straight line so that, when x m from an origin, its acceleration is given by $-9e^{-2x} \text{ ms}^{-2}$. Initially, it is at the origin where the velocity is 3 ms^{-1} .

- (i) Determine the velocity as a function of x in simplest form, justifying any choice you may have to make. 2
- (ii) Determine x as a function of t , where t is the number of seconds after it leaves the origin. 2
- (iii) Find the particle's velocity and acceleration 3 seconds after leaving the origin. 2

Question 6 (12 marks) (Use Writing Booklet 3)

Marks

- (a) A particle's motion is defined by the equation; $v^2 = 12 + 4x - x^2$, where x is its displacement from the origin in metres and v its velocity in ms^{-1} . Initially the particle 6 metres to the right of the origin.

- (i) Show that the particle is moving in Simple Harmonic Motion 1

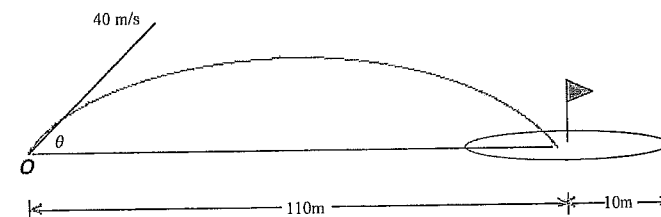
- (ii) Find the centre, period and amplitude of the motion 3

- (iii) The displacement of the particle at any time t is given by the equation. 2

$$x = a \sin(nt + \alpha) + b$$

Find the values of α and b , given $0 \leq \theta \leq 2\pi$

- (b) A golfer hits a golf ball from a point O with velocity 40 m/s at an angle θ to the horizontal. The ball travels in a vertical plane where the acceleration due to gravity is 10 ms^{-2} .



- (i) Write down expressions for the horizontal displacement x metres, and the vertical displacement y metres, of the golf ball from O after time t seconds. 1
- (ii) Hence show that the horizontal range, R metres, of the golf ball until it returns to ground level is given by $R = 160 \sin 2\theta$ 2
- (iii) The golfer is aiming over horizontal ground at a circular green of radius 10 metres, with the centre of the green 110 metres from O . Find the possible set of values of θ for the ball to land on the green, giving your answers correct to the nearest degree. 3

Question 7 (12 marks) (Use Writing Booklet 4)**Marks**

- (a) Four dice are rolled simultaneously. Any die showing a 6 on the uppermost face is set aside, and the remaining dice are rolled again.

(Note: a die has six faces numbered 1 to 6 with each face equally likely to fall uppermost)

- (i) Find the probability (correct to 2 decimal places) that after the first roll of the dice, exactly one of the four dice is showing a 6 on the uppermost face. 1
- (ii) Find the probability (correct to 2 decimal places) that after the second roll of the dice exactly two of the four dice are showing a 6 on the uppermost face. 3

- (b) (i) Show that $\frac{x^n + x^{n+2}}{1+x^2} = x^n + x^{n+1}$ 1

- (ii) An Integral is defined by $I_n = \int_0^1 \frac{x^n}{1+x^2} dx$, for $n \geq 0$

1. Evaluate I_0 1

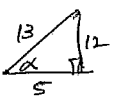
2. Use part (i) to show that $I_n + I_{n+2} = \frac{1}{n+1}$ 2

3. Evaluate I_2 1

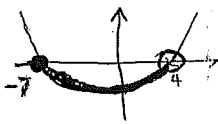
- (c) (i) Show that $\frac{\binom{3n}{k}}{\binom{3n}{k-1}} = \frac{3n-k+1}{k}$ 1

- (ii) Find the greatest coefficient in the expansion of $\left(1 + \frac{x}{2}\right)^{3n}$,
(n a positive integer). 2

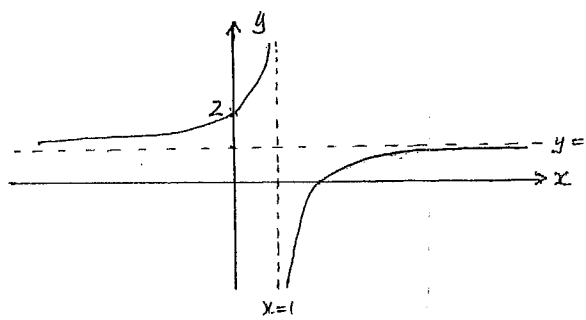
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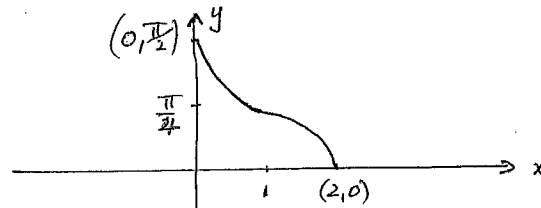
Q	Solution	Marks/Comments
1a)	$y = \tan^{-1}\left(\frac{x}{2}\right)$ $\frac{dy}{dx} = \frac{1}{1 + \left(\frac{x}{2}\right)^2} \cdot \frac{1}{2}$ $= \frac{4}{4 + x^2} \cdot \frac{1}{2}$ $= \frac{2}{4 + x^2}$	1
b)	 $\cos \alpha = \frac{5}{13}$ $\cos 2\alpha = 2\cos^2 \alpha - 1$ or use other definitions $= 2\left(\frac{5}{13}\right)^2 - 1$ of $\cos 2\alpha$ $= -\frac{119}{169}$	1 1
c)	$x^3 - 7x - 6 = 0$ $x^3 + 0x^2 - 7x - 6 = 0$ (-) (+) (-) let roots be $-1, 3, \alpha$ $\therefore -1 + 3 + \alpha = 0$ (or $-3\alpha = 6$) $\therefore \alpha = -2$ $\alpha = -2$	1
d)	$\frac{d}{dx}(x^x e^{-x^2})$ using product rule $u = x^x$ $v = e^{-x^2}$ $u' = 2x$ $v' = -2xe^{-x^2}$ $\therefore \frac{d}{dx}(x^x e^{-x^2}) = \underline{(2xe^{-x^2}) + (-2x^3e^{-x^2})}$ [use $vu' + uv'$] $= 2xe^{-x^2}(1 - x^2)$ [or $uv' + vu'$] $\text{OR} = \frac{2x(1 - x^2)}{e^{x^2}}$	1

Q	Solution	Marks/Comments
1e)	(i) $\tan 45^\circ = \left \frac{m+2-m}{1+m(m+2)} \right $ $\therefore 1 = \left \frac{2}{m^2 + 2m + 1} \right $	1
(ii)	$\frac{2}{m^2 + 2m + 1} = \pm 1$ $m^2 + 2m + 1 = 2$ or $-m^2 - 2m - 1 = 2$ $m^2 + 2m - 1 = 0$ $\therefore m^2 + 2m + 1 = -2$ $m = \frac{-2 \pm \sqrt{8}}{2}$ $m^2 + 2m + 3 = 0$ $\therefore m = -1 \pm \sqrt{2}$ $b^2 - 4ac < 0 \therefore$ no sols or $m = 0.41$ (to 2 d.p.)	1 -1 for not showing (case 2)
f)	$I = \int_1^e \frac{1}{x} (1 + \ln x)^3 dx$ $u = 1 + \ln x$ $du = \frac{1}{x} dx$ for $x = e$ $u = 2$ $x = 1$ $u = 1$	1
	$\therefore I = \int_1^2 u^3 du$ $= \left[\frac{u^4}{4} \right]_1^2$ $= \frac{16}{4} - \frac{1}{4}$ $= 3\frac{3}{4}$	1

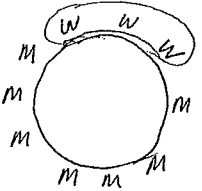
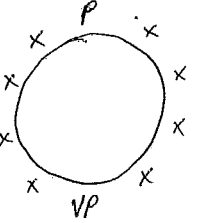
Q	Solution	Marks/Comments
2a)	$P(t+1, 2t^2+1)$ $\therefore x = t+1 \text{ --- (1)} \quad y = 2t^2+1 \text{ --- (2)}$ from (1) $t = x-1$ sub in (2) $y = 2(x-1)^2+1$ $= 2(x^2-2x+1)+1$ $= 2x^2-4x+3$	1
b)	$\frac{2x+3}{x-4} \leq 1 \quad x \neq 4$ x by $(x-4)^2$ $(2x+3)(x-4) \leq (x-4)^2$ $2x^2-5x-12 \leq x^2-8x+16$ $x^2+3x-28 \leq 0$ $(x-4)(x+7) \leq 0$ $-7 \leq x < 4$ 	$\frac{1}{2}$ off for $x \leq 4$
c)	$\lim_{x \rightarrow 0} \frac{\sin 3x}{5x} = \lim_{x \rightarrow 0} \left(\frac{3}{5} \cdot \frac{\sin 3x}{3x} \right)$ $= \frac{3}{5} \lim_{x \rightarrow 0} \frac{\sin 3x}{3x}$ $= \frac{3}{5}$	1
d)	$I = \int_0^\pi 2 \cos^2 \frac{x}{4} dx \quad \text{Note } \cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$ $\therefore \cos^2 \frac{x}{4} = \frac{1}{2} \left(1 + \cos \frac{x}{2} \right)$ $\therefore I = \int_0^\pi \left(1 + \cos \frac{x}{2} \right) dx$ $= \left[x + 2 \sin \frac{x}{2} \right]_0^\pi$ $= \pi + 2$ $\frac{d}{d\theta} \cos^2 \theta = \cos 2\theta + 1$ $\therefore 2 \cos^2 \frac{x}{4} = \cos \left(2 \cdot \frac{x}{4} \right) + 1$ $= \cos \frac{x}{2} + 1$	1

Q	Solution	Marks/Comments
2e)	Note $T_{k+1} = {}^nC_k a^{n-k} b^k$ for $(x^2 + \frac{2}{x})^{10}$ $\therefore T_{k+1} = {}^{10}C_k (x^2)^{10-k} \left(\frac{2}{x}\right)^k$ $= {}^{10}C_k x^{20-2k} \cdot \frac{2^k}{x^k}$ $= {}^{10}C_k x^{20-3k} \cdot 2^k$ $\therefore 20-3k = 5$ $15 = 3k$ $5 = k$ $\therefore T_{k+1} = {}^{10}C_5 (x^2)^5 \left(\frac{2}{x}\right)^5$ $\therefore$ Coefficient is ${}^{10}C_5 \cdot 2^5 (= 8064)$	1 $\frac{1}{2}$ $\frac{1}{2}$
f)	$A(-3, 4) \quad B(1, 2) \quad (x, y)$ $\left(\frac{nx_1 + mx_2}{m+n}, \frac{ny_1 + my_2}{m+n}\right) - 1$ $\therefore P\left(\frac{3+3}{2}, \frac{-4+6}{2}\right)$ $\therefore P(3, 1)$ check (optional)	1 1

Q	Solution	Marks/Comments
3a)	$f(x) = \frac{x-2}{x-1} \quad (x \neq 1)$	
(i)	$f'(x) = \frac{(x-1) - (x-2)}{(x-1)^2}$ $= \frac{1}{(x-1)^2} > 0 \text{ for all } x \neq 1$ Since $(x-1)^2 > 0$	1
(ii)		2
(iii)	let $y = \frac{x-2}{x-1} \quad (= f(x))$ inverse is $x = \frac{y-2}{y-1}$ $xy - x = y - 2$ $xy - y = x - 2$ $y(x-1) = x-2$ $y = \frac{x-2}{x-1}$ $\therefore f^{-1}(x) = \frac{x-2}{x-1}$	1
(iv)	the function is the inverse of itself $\therefore$ the function is symmetrical about $y=x$	1

Q	Solution	Marks/Comments
3b)	$y = \frac{1}{2} \cos^{-1}(x-1)$	
(i)	$D_f: -1 \leq x-1 \leq 1$ $0 \leq x \leq 2$ $R_f: 0 \leq y \leq \frac{\pi}{2}$ Note: $0 \leq \cos^{-1}(x-1) \leq \pi$ $\therefore 0 \leq \frac{1}{2} \cos^{-1}(x-1) \leq \frac{\pi}{2}$	1
(ii)		1
(iii)	$y = \frac{1}{2} \cos^{-1}(x-1)$ $\therefore \cos^{-1}(x-1) = 2y$ $x-1 = \cos 2y$ $x = 1 + \cos 2y$ $V = \pi \int_0^{\pi/2} (1 + \cos 2y)^2 dy$ $= \pi \int_0^{\pi/2} (1 + 2\cos 2y + \cos^2 2y) dy$ $= \pi \int_0^{\pi/2} \left[1 + 2\cos 2y + \frac{1}{2}(1 + \cos 4y) \right] dy$ $= \pi \int_0^{\pi/2} \left(\frac{3}{2} + 2\cos 2y + \frac{1}{2}\cos 4y \right) dy$ $= \pi \left[\frac{3y}{2} + \sin 2y + \frac{1}{8}\sin 4y \right]_0^{\pi/2}$ $= \pi \left[\left(\frac{3\pi}{4} + 0 + 0 \right) - (0 + 0 + 0) \right]$ $= \frac{3}{4}\pi^2 \text{ Units}^3$	1

Q	Solution	Marks/Comments
4a)		
(i)	$f(x) = 4e^{-x} - \tan x + 1$ $f(1) = 0.91411004 > 0$ $f(1.5) = -12.20889 < 0 \quad \therefore 1 < x < 1.5$	1
(ii)	now $f'(x) = -4e^{-x} - \sec^2 x$ $\therefore f'(1) = -4.897036585$ $x_2 = 1 - \frac{f(1)}{f'(1)}$ $x_2 = 1 + \frac{0.91411004}{4.897036585}$ $\therefore x_2 = 1.187 \text{ (4 sig. figs)}$ Note: $f(1.187) = -0.255801167$ which is closer to zero than $f(1)$ $\therefore$ is a better approximation	1
b)		
(i)	$N = 5000 + Ae^{kt} \quad \text{--- (1)}$ from (1) $Ae^{kt} = N - 5000$ Now $\frac{dN}{dt} = Ae^{kt} \cdot k$ $= k(N - 5000)$	1
(ii)	when $t=0 \quad N=15000$ $\therefore 15000 = 5000 + Ae^0$ $\therefore A = 10000$ $\therefore N = 5000 + 10000e^{kt}$ when $t=2 \quad N=20000$ $\therefore 20000 = 5000 + 10000e^{2k}$ $\therefore e^{2k} = \frac{15000}{10000} = 1.5$ $\therefore 2k = \ln 1.5$ $\therefore k = \frac{\ln 1.5}{2} \doteq 0.20273$	1

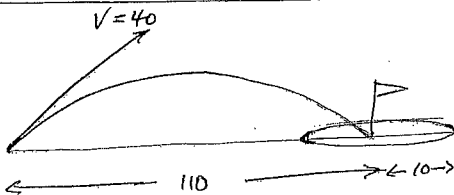
Q	Solution	Marks/Comments
4b)		
(iii)	when $t=7$ $N = 5000 + 10000e^{7 \times \frac{\ln 1.5}{2}}$ $= 46335 \text{ (nearest whole number)}$	1
c)		
(i)	 If the three women are to sit together there are essentially 8 items to arrange around the table $\therefore P(3 \text{ women sit together}) = \frac{7!3!}{9!}$ $= \frac{1}{12}$	1
(ii)	 $P(P \text{ \& VP sit opp}) = \frac{8!}{9!}$ $= \frac{1}{9}$ Note: Once P & VP have been seated opposite one another there are 8 seats left to fill $\therefore 8!$	1

Q	Solution	Marks/Comments
5a)	Prove. $\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{(n-1)}{n!} = 1 - \frac{1}{n!} \quad n \geq 2$ Step 1 Prove true for $n=2$ LHS = $\frac{1}{2!} = \frac{1}{2}$ RHS = $1 - \frac{1}{2!} = \frac{1}{2}$ $\therefore$ true for $n=2$ Step 2 Assume true for $n=k$ ($2 \leq k < n$) i.e. $\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{k-1}{k!} = 1 - \frac{1}{k!}$ Step 3 Aim to prove true for $n=k+1$ i.e. A.T.P. $\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{k-1}{k!} + \frac{k}{(k+1)!} = 1 - \frac{1}{(k+1)!}$ L.H.S. = $1 - \frac{1}{k!} + \frac{k}{(k+1)!}$ [L.C.D. = $(k+1)!]$ = $\frac{(k+1)! - (k+1) + k}{(k+1)!}$ = $\frac{(k+1)! - 1}{(k+1)!}$ = $1 - \frac{1}{(k+1)!}$ = R.H.S. Step 4 1. true for $n=k+1$ if true for $n=k$ Since true for $n=2$ then true for $n=3, 4, \dots$ $\therefore$ by principle of mathematical induction true for $n \geq 2$ (n integer)	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ 1 1 $\frac{1}{2}$

Q	Solution	Marks/Comments
5b)	$\frac{d^2x}{dt^2} = \frac{d}{dt} \cdot \frac{dx}{dt}$ = $\frac{dv}{dt}$ = $\frac{dv}{dx} \times \frac{dx}{dt}$ = $v \cdot \frac{dv}{dx}$ = $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) \times \frac{dx}{dx}$ = $\frac{d}{dx} \left(\frac{1}{2} v^2 \right)$	1
c)	Using part (b) $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -9e^{-2x}$ $\frac{1}{2} v^2 = -9 \int e^{-2x} dx$ $\frac{1}{2} v^2 = \frac{9}{2} e^{-2x} + C$ when $x=0 \quad v=3$ $\therefore \frac{9}{2} = \frac{9}{2} + C$ $\therefore C = 0$ $\therefore \frac{1}{2} v^2 = \frac{9}{2} e^{-2x}$ $v^2 = 9e^{-2x}$ $v = \pm \sqrt{9e^{-2x}}$ when $x=0 \quad v=3 \quad \therefore v = \sqrt{9e^{-2x}}$ $\therefore v = 3e^{-x}$ Could also justify the positive choice because initially velocity is positive and v never equals zero $\therefore$ must continue positive direction	1 $\frac{1}{2}$

Q	Solution	Marks/Comments
Q5c (ii)	$\frac{dx}{dt} = 3e^{-x} \text{ from (i)}$ $\frac{dt}{dx} = \frac{1}{3e^{-x}} = \frac{1}{3}e^x$ $t = \frac{1}{3} \int e^x dx$ $= \frac{1}{3}e^x + C$ When $t=0$ $x=0$ $\therefore 0 = \frac{1}{3} + C$ $\therefore C = -\frac{1}{3}$ $\therefore t = \frac{1}{3}e^x - \frac{1}{3}$ $3t = e^x - 1$ $e^x = 3t + 1$ $\therefore x = \ln(3t + 1)$	1
(iii)	or when $t=3$ $x = \ln 10$ $v = \frac{dx}{dt}$ $\therefore v = \frac{3}{3t+1}$ $= \frac{3}{9+1}$ $= 0.3 \text{ ms}^{-1}$ $a = \frac{dv}{dt}$ $= -3(3t+1)^{-2} \cdot 3$ $= -\frac{9}{(3t+1)^2}$ $= -\frac{9}{(9+1)^2}$ $= -0.09 \text{ ms}^{-2}$	1
	$\therefore v = 3e^{-\ln 10}$ $= 3e^{\ln \frac{1}{10}}$ $= 3 \times \frac{1}{10}$ $= 0.3 \text{ m/s}$ $a = -9e^{-2 \ln 10}$ $= -9e^{\ln \frac{1}{100}}$ $= -9 \cdot \frac{1}{100}$ $= -0.09 \text{ m/s}^2$	1

Q	Solution	Marks/Comments
Q6 a) (i)	$v^2 = 12 + 4x - x^2$ now $a = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$ $\ddot{x} = \frac{d}{dx} \left(6 + 2x - \frac{x^2}{2} \right)$ $= 2 - x$ $= -1(x-2) \therefore \text{SHM. as } a = -n^2(x-k)$	1
(ii)	Centre of motion is $x=2$ where $\ddot{x}=0$ $n=1 \therefore \text{Period } T = \frac{2\pi}{n} = 2\pi$ Extremities of motion where $v=0$ $\therefore 12 + 4x - x^2 = 0$ $(2+x)(6-x) = 0$ $\therefore x = -2, 6$ $\therefore \text{amplitude} = 4$	1
(iii)	$a=4$ $n=1$ and centre of motion is $k=2$ $\therefore x = 4 \sin(t+\theta) + 2$ When $t=0$ $x=6$ (given) $\therefore 6 = 4 \sin \theta + 2$ $\therefore \sin \theta = 1$ $\theta = \frac{\pi}{2}$ $\therefore x = 4 \sin\left(t + \frac{\pi}{2}\right) + 2$	1

Q	Solution	Marks/Comments
6.1)		
(i)	$x = 40t \cos \theta$ $y = 40t \sin \theta - 5t^2$	1
(ii)	When ball returns to horizontal $y = 0$ $\therefore 40t \sin \theta - 5t^2 = 0$ $5t(8 \sin \theta - t) = 0$ $t = 0, 8 \sin \theta$ When $t = 8 \sin \theta$ $x = 320 \sin \theta \cos \theta$ $= 160 \sin 2\theta$ (Note: $\sin 2\theta = 2 \sin \theta \cos \theta$)	1
(iii)	$x = 100$ $160 \sin 2\theta = 100$ $\sin 2\theta = \frac{5}{8}$ $2\theta = 39^\circ, 141^\circ$ (nearest degree) $\theta = 19^\circ, 71^\circ$ (nearest degree) $x = 120$ $160 \sin 2\theta = 120$ $\sin 2\theta = \frac{3}{4}$ $2\theta = 49^\circ, 131^\circ$ (nearest deg) $\theta = 24^\circ, 66^\circ$ (nearest deg) $\therefore 19^\circ < \theta < 24^\circ$ OR $66^\circ < \theta < 71^\circ$	1

Q	Solution	Marks/Comments
Q7a)		
(i)	$P(\text{one six on first roll}) = {}^4C_1 \left(\frac{5}{6}\right)^3 \left(\frac{1}{6}\right)^1$ $= 0.39$ (2 dp)	1
(ii)	$P(2 \text{ sixes on first roll and no 6's on second roll})$ $= {}^4C_2 \left(\frac{5}{6}\right)^2 \left(\frac{1}{6}\right)^2 \times {}^2C_0 \left(\frac{5}{6}\right)^2 \left(\frac{1}{6}\right)^0 = 0.0804$ $P(1 \text{ six on first roll and one 6 on second roll})$ $= {}^4C_1 \left(\frac{5}{6}\right)^3 \left(\frac{1}{6}\right)^1 \times {}^3C_1 \left(\frac{5}{6}\right)^2 \left(\frac{1}{6}\right)^1 = 0.1340$ $P(\text{no 6's on first roll and two 6's on second roll})$ $= {}^4C_0 \left(\frac{5}{6}\right)^4 \left(\frac{1}{6}\right)^0 \times {}^4C_2 \left(\frac{5}{6}\right)^2 \left(\frac{1}{6}\right)^2 = 0.0558$ $\therefore P(\text{two sixes overall}) = 0.0804 + 0.1340 + 0.0558$ $= 0.27$ (2 dp)	1
b.1)	$\frac{x^n + x^{n+2}}{1 + x^2} = \frac{x^n(1 + x^2)}{1 + x^2}$ $= x^n$	1
ii)	$I_n = \int_0^1 \frac{x^n}{1+x^2} dx$ $I_0 = \int_0^1 \frac{1}{1+x^2} dx$ $= [\tan^{-1} x]_0^1$ $= \frac{\pi}{4}$	1

Q	Solution	Marks/Comments
Q7b (11)		
2.	$I_n = \int_0^1 \frac{x^n}{1+x^2} dx \quad I_{n+2} = \int_0^1 \frac{x^{n+2}}{1+x^2} dx$ $\therefore I_n + I_{n+2} = \int_0^1 \frac{x^n}{1+x^2} dx + \int_0^1 \frac{x^{n+2}}{1+x^2} dx$ $= \int_0^1 \frac{x^n + x^{n+2}}{1+x^2} dx$ $= \int_0^1 x^n dx \quad \text{from part (i)}$ $= \left[\frac{x^{n+1}}{n+1} \right]_0^1$ $= \frac{1}{n+1}$	1
3.	from 2. put $n=0$ $\therefore I_0 + I_2 = \frac{1}{0+1}$ $\therefore I_2 = 1 - I_0$ $= 1 - \frac{\pi}{4}$ OR $I_2 = \int_0^1 \frac{x^2}{1+x^2} dx$ $= \int_0^1 \frac{1+x^2-1}{1+x^2} dx$ $= \int_0^1 1 dx - \int_0^1 \frac{1}{1+x^2} dx$ $= [x]_0^1 - [\tan^{-1}x]_0^1$ $= 1 - \frac{\pi}{4}$	1
	OR (1)	

Q	Solution	Marks/Comments
Q7c)		
(i)	$\frac{\binom{3n}{k}}{\binom{3n}{k-1}} = \frac{\frac{3n!}{(3n-k)!k!}}{\frac{3n!}{(3n-k+1)!(k-1)!}}$ $= \frac{3n-k+1}{k}$	1
(ii)	$\frac{T_{k+1}}{T_k} = \frac{\binom{3n}{k} a^{3n-k} b^k}{\binom{3n}{k-1} a^{3n-k+1} b^{k-1}}$ $= \frac{\binom{3n}{k}}{\binom{3n}{k-1}} \cdot \frac{b}{a}$ $= \frac{3n-k+1}{k} \cdot \frac{1}{2} \quad \text{from (i)}$ $= \frac{3n-k+1}{2k}$ Now $\frac{3n-k+1}{2k} > 1$ for increasing coefficients $\therefore 3n-k+1 > 2k \quad (k>0)$ $\therefore 3k < 3n+1$ $k < n + \frac{1}{3}$ $\therefore k = n \text{ for greatest coefficient}$ $\therefore$ greatest Coefficient $= {}^{3n}C_n \left(\frac{1}{2}\right)^n$ [Note: $T_{k+1} = {}^{3n}C_k a^{3n-k} b^k$] $a=1$ $b=\frac{1}{2}$	1